Matvei Borodin

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Summary

ML Engineer and AI Researcher focused on LLM agents, reinforcement learning, and efficient inference. Experienced in building RL training pipelines, evaluation frameworks, and production-oriented ML systems.

Experience

AI Researcher, AIRI *Moscow, Russia*
LLM Agents Research Group *September 2026 — Present*

- Researching **adaptive evolution of verified tool-use environments** for reinforcement learning of **small LLM agents** (Qwen3.5 2B–4B).
- Working on a custom **DeepSeek-based agent harness** for scalable environment generation, deterministic verification, and student-adaptive difficulty calibration.
- Investigating **adaptive RLVR/GRPO training** and autonomous feedback loops for **Recursive Self-Improvement (RSI)**.

Alice Voice Activation Team *Moscow, Russia*
ML Engineer Intern, Yandex *May 2026 — August 2026*

- Developed a compact **hybrid CNN–Transformer architecture** for on-device **keyword spotting** and implemented **FlashAttention** with **QKNorm**, **relative position biases**, and **RoPE**.
- Built an end-to-end converter from **FP32 PyTorch** to fully quantized **TFLite INT8** without quality degradation, and implemented **KV-cached streaming inference** for on-device deployment.
- Achieved a **15% relative improvement** in target **quality metrics**, reduced compute, and kept **memory usage and inference latency** within production budgets.
- Developed an **embedded C++ orchestration layer** to centrally manage **on-device models** and coordinate their activation.

Alignment Team for Alice Search Generative Answers *Moscow, Russia*
ML Engineer Intern, Yandex *December 2025 — March 2026*

- Implemented and launched a **GRPO training pipeline** for the Alice Search **RAG agent**, including **online reward computation** and core alignment and evaluation components.
- Ran experiments with **CERL, DPO, and GRPO**, analyzing training-data composition and hyperparameters; diagnosed **reward hacking** and quality regressions through assessor evaluations.

MultiHopBench: A Constraint-Aware Benchmark for Factual QA *Moscow, Russia*
YandexGPT Pretraining Analytics, YSDA (School of Data Analysis) *October 2025 — June 2026*

- Designed a bilingual Russian–English benchmark with **1,300 factual QA examples** for multi-hop retrieval, constraint satisfaction, and no-answer behavior; built a reproducible **Wikidata/SPARQL pipeline** and an evaluation framework combining an **LLM judge** with a **deterministic scorer**.

Education

ITMO University, AI Talent Hub *Saint Petersburg, Russia*
Master's degree in Artificial Intelligence *2026 — Present*

Higher School of Economics *Moscow, Russia*
Bachelor's degree, Faculty of Computer Science *2022 — 2026*

Technical Skills

Languages: Python, C++, Swift, Kotlin, Java
Technologies: PyTorch, SQL, Nirvana (yandex apache airflow analog), Transformers, vLLM
Skills: Deep Learning, NLP, LLM, RL, RAG, Speech, LoRA/PEFT, model quantization